Sahaj Bhadja

(213) 284-4651 sahajbhadja10@gmail.com sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with 2+ years building **machine-learning and image-processing systems**, numerical evaluation pipelines, and concurrent production services using **Python, C++, PyTorch, computer vision**, automated testing, and CI/CD.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Advanced NLP, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Reworked **fault-tolerant asynchronous services** with explicit job states, isolated failures, and production monitoring, tripling throughput for 1K+ concurrent jobs while sustaining 99%+ reliability across 5K+ monthly runs.
- Optimized data paths and indexed queries across PostgreSQL and SQL Server, reducing dashboard latency from 1.2 seconds to 350 milliseconds while supporting 500+ users through **measured performance improvements**.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built **multithreaded image-ingestion workflows** for PDF, DOCX, ZIP, OCR, and vision-model inputs, processing 1K+ files per batch while reliably reducing instructor feedback turnaround by 60% in production.
- Integrated multimodal models, OCR, and automatic fallbacks into repeatable evaluation workflows, serving 10K+ daily requests while validating changes through **unit testing, debugging, and CI/CD** safely in production.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a Decision Transformer for autonomous-driving offline reinforcement learning, improving trajectory planning 18% while processing 100K+ simulated driving logs through reproducible **Python evaluation pipelines** for controlled experiments.
- Implemented a spatiotemporal encoder integrating HD-map geometry, lane context, and ego trajectories with multi-head attention, consistently improving closed-loop stability 15% across **measured simulation baselines** for planning research.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Designed cross-platform integration modules using **object-oriented architecture**, reducing API response times 30%, technical debt 20%, and duplicated logic across a highly reliable, scalable, maintainable microservice-based data-sharing platform.
- Automated 20+ **unit and integration tests** with PyTest, Mockito, Jenkins, and virtual-machine builds across Windows, Linux, and macOS, collaborating effectively through Agile reviews and Git workflows worldwide.

Cyber Security and Data Mining **May 2023 – August 2023**
Data Analyst – Research Fellowship *India*

- Engineered statistical and behavioral features across 650K+ URLs, comparing Random Forest, recurrent, convolutional, and bidirectional models through **reproducible machine-learning experiments** for accurate malicious-link classification across diverse datasets.
- Achieved **96%+ detection accuracy** using 21 behavioral indicators and sequence models, analyzing prediction patterns to identify phishing and malware signals while documenting experimental results and model tradeoffs.

Skills

Languages & Numerical: Python, C++, Java, C, SQL, JavaScript, Bash, Shell, NumPy, Linear Algebra, Numerical Computing
Vision & Machine Learning: OpenCV, PyTorch, TensorFlow, Image Processing, Computer Vision, Image Segmentation, Classification, Detection, OCR, CNNs, Transformers, Feature Engineering
Systems: Multithreading, Concurrency, Asynchronous Programming, Linux, Windows, Memory Management, Performance Profiling, Hardware-Software Integration, Real-Time Constraints, TCP/IP
Data & Platforms: PostgreSQL, SQL Server, MySQL, MongoDB, Redis, REST APIs, Event-Driven Systems, Docker, Kubernetes, AWS, Microsoft Azure
Engineering: Git, GitHub Actions, Jenkins, CI/CD, PyTest, Mockito, Unit Testing, Integration Testing, Debugging, Agile, Design Reviews, Code Reviews, Requirements Analysis, System Design, Technical Documentation, AI-Assisted Development

Projects

VisionInspect Pipeline | *Python, C++, OpenCV, PyTorch, NumPy, Docker*

- Designed a **machine-vision inspection pipeline** for image calibration, normalization, segmentation, defect classification, and review overlays, separating acquisition and inference stages while validating algorithms against synthetic illumination, noise, and distortion conditions.

OptiMotion Diagnostics | *C++, Python, Multithreading, OpenCV, TCP/IP, PostgreSQL*

- Built a **cross-subsystem diagnostic simulator** connecting camera acquisition, motion states, telemetry, and algorithm outputs, with multithreaded queues, fault injection, synchronized event traces, and repeatable tests for root-cause analysis.

PiXMod | *Python, PyTorch, CLIP, SAM, Stable Diffusion, MongoDB*

- Built an **AI-powered image-processing platform** using SAM segmentation, Stable Diffusion restoration, and CLIP semantic retrieval, processing 1K+ edited assets while maintaining scalable image and metadata storage in MongoDB.